

STAT8561 - Linear Statistical Analysis I

2026 Fall Lecture Notes

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Preface

The topic of this course includes statistical inference, Multivariate normal distribution, distribution of quadratic forms, linear models, regression models and experimental design models.

Prerequisites

Math 4751/6751 Mathematical Statistics I and Math 4752/6752 Mathematics and Statistics II.

Instructor

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Office Hour

9:00 - 12:00 on Wednesday or by appointment

Grade Distribution

- Homework – 40%
- Exam 1 – 20%
- Exam 2 – 20%
- Final – 20%
- Attendance – *7%

Assignment

- A1, TBA

Exam

- TBA

Topics and Corresponding Lectures

Those chapters are based on the lecture notes. This part will be updated frequently.

Status	Chapter	Topic	Lecture
		Welcome and Overview	1
	Ch. 1	Math Notation	2–
	Ch. 2	Least Squares Estimation	TBA

Recommended Textbooks

- Ding, Peng. *Linear Model and Extensions*. CRC Press, 2027. [arXiv link](#)
- Montgomery, Douglas C., Elizabeth A. Peck, and G. Geoffrey Vining. *Introduction to Linear Regression Analysis*. John Wiley & Sons, 2021. (Montgomery et al. 2021)
- Seber, George AF, and Alan J. Lee. *Linear Regression Analysis*. John Wiley & Sons, 2003. (Seber and Lee 2003)

Side Readings

- Montgomery, Douglas C. (2017). *Design and Analysis of Experiments*. John Wiley & Sons. (Montgomery 2017)

Introduction

This is an introduction to the course for linear statistical analysis. It will give you an overview of what we will be covering in the course and how to get the most out of it. We will be covering a wide range of topics in linear statistical analysis,

- simple linear regression
- multiple linear regression
- generalized linear models
- mixed effects models
- quantile regression.

We will also be discussing the assumptions underlying these models and how to check them.

The course will be structured around lectures, homework assignments, a midterm and a final.

To get the most out of this course, it is important to attend all lectures and complete all homework assignments. It is also important to ask questions and participate in class discussions. The more you engage with the material, the more you will learn.

I am looking forward to a great semester and I hope you are too!

To illustrate the concepts we will be covering in this course, let's consider a simple example. Suppose we have a data set of heights and weights of individuals. We want to understand the relationship between height and weight, and we can use linear regression to model this relationship.

```
# Load necessary libraries
library(ggplot2)
library(dplyr)
# Create a sample data set

set.seed(8561)

theme_set(theme_minimal())

n <- 100
heights <- rnorm(n, mean = 170, sd = 10)
weights <- 0.5 * heights + rnorm(n, mean = 0, sd
```

```
= 5)
df <- data.frame(heights, weights)
head(df, 5)
```

```
      heights  weights
1 179.2812 88.01599
2 168.0474 76.10017
3 178.3029 83.46016
4 178.7006 82.01763
5 172.6678 92.84592
```

```
# Fit a linear regression model
model <- lm(weights ~ heights, data = df)
# Summarize the model
summary(model)
```

```
Call:
lm(formula = weights ~ heights, data = df)
```

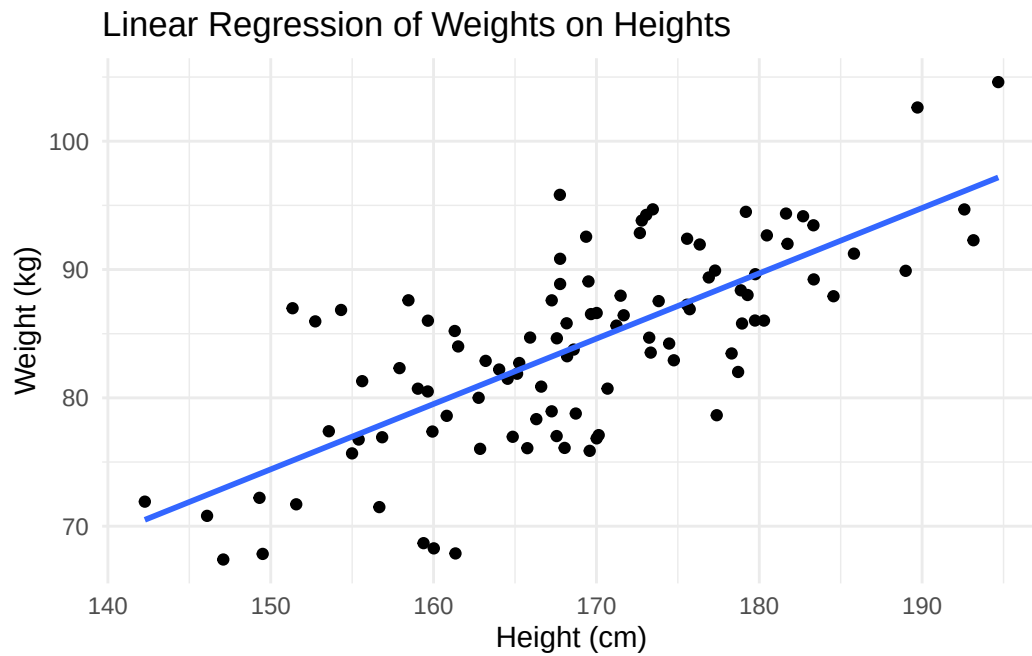
```
Residuals:
      Min       1Q   Median       3Q      Max
-12.3303  -3.8998  -0.2191   3.2380  12.3478
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.92852     8.22226  -0.235   0.815
heights      0.50907     0.04857  10.481 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 5.251 on 98 degrees of freedom
Multiple R-squared:  0.5285,    Adjusted R-squared:  0.5237
F-statistic: 109.9 on 1 and 98 DF,  p-value: < 2.2e-16
```

```
# Plot the data and the fitted line
ggplot(df, aes(x = heights, y = weights)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Linear Regression of Weights on Heights",
       x = "Height (cm)",
       y = "Weight (kg)")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



In this example, we generated a data set of heights and weights, fitted a linear regression model to the data, and visualized the relationship between height and weight. This is just a simple example, but it illustrates the types of analyses we will be doing in this course. We will be covering much more complex models and data sets as we progress through the semester.

Throughout the course, we will be using **R** programming software.

1 Matrix Representation and Notation

In this first week, we introduce the mathematical language and statistical framework that will be used throughout the course. Our focus is on the notation of vectors and matrices, vectors of random variables, expectation and covariance operators, and the basic form of the linear regression model.

Learning Objectives

By the end of this week, students should be able to:

- use standard matrix notation for linear statistical models;
- distinguish between scalars, vectors, matrices, random variables, and random vectors;
- compute expectations and covariance matrices for random vectors;
- interpret the linear regression model in matrix form;
- understand why projection and least squares will play a central role in this course.

Reading

Recommended reading for this week:

- Seber and Lee (2003), Chapter 1:
 - 1.1 Notation
 - 1.2 Statistical Models
 - 1.3 Linear Regression Models
 - 1.4 Expectation and Covariance Operators
- Optional preview:
 - Chapter 2: Multivariate Normal Distribution

1.1 Why Linear Statistical Analysis?

Linear statistical analysis is one of the central foundations of graduate statistics. Many methods that at first look different are built on the same underlying structure:

- regression
- analysis of variance
- analysis of covariance
- prediction
- model comparison
- and parts of generalized linear modelling

A major goal of this course is to see these topics under a unified framework.

1.2 A unifying point of view

A large part of the course can be summarized by the model

$$Y = X\beta + \varepsilon,$$

where

- Y is a response vector,
- X is a design matrix,
- β is an unknown parameter vector,
- ε is a random error vector.

This compact expression contains a great deal of statistical structure. Over the semester, we will study how to estimate β , quantify uncertainty, test hypotheses, diagnose model failures, and make predictions.

1.3 Basic Notation

Scalars, vectors, and matrices

We use the following conventions throughout the course:

- scalars are written in lowercase italic letters, such as a , b , n ;
- vectors are written in bold lowercase letters, such as $\mathbf{x}$, $\mathbf{y}$;
- matrices are written in bold uppercase letters, such as $\mathbf{X}$, $\mathbf{A}$;
- random variables are often written in uppercase letters, such as Y ;
- realizations of random variables are written in lowercase letters, such as y .

Univariate case

A random variable is **univariate** if it takes values in one dimension. We write

$$X \in \mathbb{R}.$$

A realization, or observed value, of X is denoted by x .

Some common quantities associated with a univariate random variable X include:

- **Mean**

$$\mu = \mathbb{E}(X).$$

- **Variance**

$$\sigma^2 = \text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

- **k th moment**

$$\mathbb{E}(X^k).$$

- **Cumulative distribution function (CDF)**

$$F(x) = P(X \leq x).$$

- **Probability density function (PDF)**, for a continuous random variable,

$$f(x).$$

- **Probability mass function (PMF)**, for a discrete random variable,

$$p(x) = P(X = x).$$

- **Quantile**

$$q_\alpha = F^{-1}(\alpha),$$

where $0 < \alpha < 1$.

For a random sample of size n , we write

$$X_1, X_2, \dots, X_n,$$

with observed values

$$x_1, x_2, \dots, x_n.$$

The sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i,$$

and the sample variance is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Multivariate case

A column vector in $\mathbb{R}^n$ is written as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = (x_1, x_2, \dots, x_n).$$

An $n \times p$ matrix is written as

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}.$$

Transpose, inverse, and rank

If $\mathbf{A}$ is a matrix, then:

- $\mathbf{A}^\top$ denotes its transpose;
- $\mathbf{A}^{-1}$ denotes its inverse, when it exists;
- $\text{rank}(\mathbf{A})$ denotes its rank;
- $\mathbf{I}_n$ denotes the $n \times n$ identity matrix.

1.3.1 Inner product and norm

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the inner product is

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i.$$

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the inner product is

$$\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}.$$

These ideas are fundamental because least squares estimation is based on minimizing squared Euclidean distance.

A random vector is a vector whose entries are random variables. For example,

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}$$

is an n -dimensional random vector.

In linear models, the response is naturally treated as a random vector.

The mean vector of $\mathbf{Y}$ is

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} \mathbb{E}[Y_1] \\ \mathbb{E}[Y_2] \\ \vdots \\ \mathbb{E}[Y_n] \end{pmatrix}.$$

We often write $\mathbb{E}[\mathbf{Y}] = \boldsymbol{\mu}$.

The covariance matrix of $\mathbf{Y}$ is

$$\text{Var}(\mathbf{Y}) = \mathbb{E}[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top].$$

Equivalently,

$$\text{Var}(\mathbf{Y}) = \mathbb{E}[\mathbf{Y}\mathbf{Y}^\top] - \boldsymbol{\mu}\boldsymbol{\mu}^\top.$$

Suppose $\mathbf{Y} = (Y_1, Y_2, Y_3)$.

Then the corresponding covariance matrix is

$$\text{Var}(\mathbf{Y}) = \begin{pmatrix} \text{Var}(Y_1) & \text{Cov}(Y_1, Y_2) & \text{Cov}(Y_1, Y_3) \\ \text{Cov}(Y_2, Y_1) & \text{Var}(Y_2) & \text{Cov}(Y_2, Y_3) \\ \text{Cov}(Y_3, Y_1) & \text{Cov}(Y_3, Y_2) & \text{Var}(Y_3) \end{pmatrix}.$$

1.3.2 Expectation and Covariance Operators

If $\mathbf{A}$ is a constant matrix and $\mathbf{b}$ is a constant vector, then

$$\mathbb{E}[\mathbf{A}\mathbf{Y} + \mathbf{b}] = \mathbf{A}\mathbb{E}[\mathbf{Y}] + \mathbf{b}.$$

This is one of the most useful identities in the course.

If $\mathbf{A}$ is a constant matrix, then

$$\text{Var}(\mathbf{A}\mathbf{Y}) = \mathbf{A} \text{Var}(\mathbf{Y}) \mathbf{A}^\top.$$

More generally, if $\mathbf{Y}$ and $\mathbf{Z}$ are random vectors, then

$$\text{Cov}(\mathbf{A}\mathbf{Y}, \mathbf{B}\mathbf{Z}) = \mathbf{A} \text{Cov}(\mathbf{Y}, \mathbf{Z}) \mathbf{B}^\top.$$

These formulas are essential for deriving the variance of estimators later.

If $Y_1, \dots, Y_n$ are independent and each has variance σ^2 , then

$$\text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

This is the most common starting assumption in classical linear regression.

1.4 Statistical Models

A statistical model is a set of probability distributions that may plausibly describe the data-generating mechanism.

1.4.1 General idea

Suppose we observe data y from a random quantity Y . A model introduces assumptions about the distribution of Y , often indexed by an unknown parameter θ .

For example:

$$Y \sim N(\mu, \sigma^2)$$

with unknown parameters μ and σ^2 .

In regression, the model is not only about the marginal distribution of the response but also about how the mean changes with explanatory variables.

1.4.2 Deterministic part and random part

A useful way to think about a statistical model is:

$$\text{data} = \text{systematic part} + \text{random part}.$$

For linear regression, this becomes

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

Here,

- $\beta_0 + \beta_1 x_i$ is the systematic part;
- ε_i is the random part.

1.5 The Linear Regression Model

Perhaps the most simple model is the *simple linear regression* where there are only **one** independent variable and **one** response variable.

A simple linear regression model is defined as

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n,$$

with the typical assumption as

$$\mathbb{E}[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2, \quad \text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \text{ for } i \neq j.$$

Matrix form

We can write the model compactly as

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

where

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

This notation will allow us to treat simple regression, multiple regression, ANOVA, and ANCOVA in one common language.

Under the same assumption,

$$\mathbb{E}[\varepsilon] = \mathbf{0} \quad \text{and} \quad \text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n,$$

we have the following.

$$\mathbb{E}[\mathbf{Y}] = \mathbf{X}\beta, \quad \text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

These are immediate consequences of the expectation and covariance rules above.

1.5.1 Geometry Preview

A central idea in linear regression is projection.

The fitted values $\hat{\mathbf{Y}}$ will later be obtained by projecting $\mathbf{Y}$ onto the column space of $\mathbf{X}$.

The column space of $\mathbf{X}$ is

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\beta : \beta \in \mathbb{R}^p\}.$$

This is the set of all mean vectors that the model can represent.

1.6 Why projection matters

The least squares estimator chooses $\hat{\beta}$ so that

$$\|\mathbf{Y} - \mathbf{X}\beta\|^2$$

is minimized.

So regression is not only algebra. It is also geometry.

Suppose we observe the following data:

$$\begin{array}{c|cccc} x_i & 0 & 1 & 2 & 3 \\ \hline y_i & 1 & 3 & 3 & 5 \end{array}$$

Then

$$\mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

We will learn next week that the least squares estimator is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

For now, the main point is to understand how the model is written and how the data are represented in matrix form.

R code

```
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

mod <- lm(y ~ x)
summary(mod)
```

```

Call:
lm(formula = y ~ x)

Residuals:
    1     2     3     4 
-0.2  0.6 -0.6  0.2 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.2000     0.5292   2.268  0.1515
x              1.2000     0.2828   4.243  0.0513 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6325 on 2 degrees of freedom
Multiple R-squared:  0.9, Adjusted R-squared:  0.85 
F-statistic:  18 on 1 and 2 DF,  p-value: 0.05132

```

```

# Inspecting the design matrix
model.matrix(mod)

```

```

      (Intercept) x
1              1 0
2              1 1
3              1 2
4              1 3
attr("assign")
[1] 0 1

```

```

# Fitted values and residuals
fitted(mod)

```

```

    1     2     3     4 
1.2 2.4 3.6 4.8

```

```

residuals(mod)

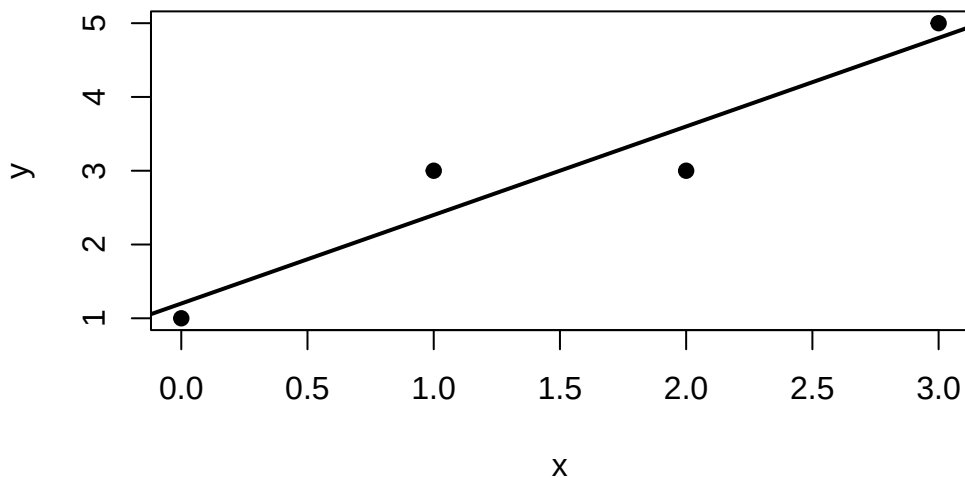
```

```

    1     2     3     4 
-0.2  0.6 -0.6  0.2

```

```
# quick plot
plot(x, y, pch = 19, xlab = "x", ylab = "y")
abline(mod, lwd = 2)
```



1.7 Discussion

1. Why is it helpful to write regression models in matrix form rather than only scalar notation?
2. What does the covariance matrix tell us that separate variances do not?
3. What is the interpretation of the column space of $\mathbf{X}$?
4. In what sense is regression a projection problem?

1.8 Practice Problems

Conceptual

1. Explain the difference between a random variable and a random vector.
2. Explain why $\text{Var}(\mathbf{Y})$ must be a symmetric matrix.

3. Give an example of a statistical model outside regression.

Computational

Let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}$$

with

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \text{Var}(\mathbf{Y}) = \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}.$$

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

Compute:

1. $\mathbb{E}[\mathbf{AY} + \mathbf{b}]$
2. $\text{Var}(\mathbf{AY})$

Regression setup

For the model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

write out the matrices $\mathbf{Y}$, $\mathbf{X}$, β , and ε for $n = 5$ observations.

1.9 Take home Practice

Complete the following:

- review matrix multiplication and transpose rules;
- derive $\mathbb{E}[\mathbf{AY} + \mathbf{b}]$ from first principles;
- derive $\text{Var}(\mathbf{AY})$ using the definition of covariance;
- write the simple linear regression model in matrix form for a dataset of your choice;
- fit a simple regression in R and report:
- the estimated coefficients,

- fitted values,
- residuals,
- and a scatterplot with the fitted line.

1.10 Summary

This week introduced the notation and basic probabilistic tools needed for the rest of the course. We defined random vectors, mean vectors, covariance matrices, and the matrix form of the linear regression model. These ideas will support everything that follows.

Next week, we will study least squares estimation and the geometry of projection in more detail.

For some optional review of the matrix algebra, see [Matrix Algebra Review](#).

2 Least Square Estimation

In this week, we study the core idea behind linear regression: **least squares estimation** (LSE). We derive the ordinary LSE, interpret it geometrically as a projection, and introduce the fitted values, residuals, and the hat matrix.

Learning Objectives

By the end of this week, students should be able to:

- define the least squares criterion for a linear model;
- derive the normal equations;
- obtain the ordinary least squares estimator when the design matrix has full column rank;
- interpret least squares as an orthogonal projection;
- define the fitted values, residuals, and hat matrix;
- explain the orthogonality properties of residuals.

Reading

Recommended reading for this week:

- Seber and Lee:
 - Chapter 3: sections on least squares estimation
- Montgomery, Peck, and Vining:
 - introductory sections on estimation in linear regression

Recap of Linear model

Recall from [Week 1](#), we discussed

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

where

- $\mathbf{Y}$ is an $n \times 1$ response vector,
- $\mathbf{X}$ is an $n \times p$ design matrix,
- β is a $p \times 1$ parameter vector,
- ε is an $n \times 1$ error vector.

Under the classical setup, we often assume

$$\mathbb{E}[\varepsilon] = \mathbf{0}, \quad \text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n.$$

Hence,

$$\mathbb{E}[\mathbf{Y}] = \mathbf{X}\beta, \quad \text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

Our goal is to estimate β from the observed data.

2.1 The Least Squares Criterion

Motivation

For any candidate value β , the model predicts $\mathbf{X}\beta$.

The discrepancy between the observed response $\mathbf{Y}$ and the model mean $\mathbf{X}\beta$ is

$$\mathbf{Y} - \mathbf{X}\beta.$$

This vector is called the **residual vector** for the candidate β .

A natural idea is to choose β so that this discrepancy is as small as possible.

2.2 Residual sum of squares

The least squares criterion is

$$S(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta).$$

Equivalently,

$$S(\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|^2.$$

We choose $\hat{\beta}$ to minimize $S(\beta)$.

We can choose different $\hat{\beta}$, but there is the best one we can find.

The least squares estimator $\hat{\beta}$ is the value of β that minimizes the residual sum of squares $S(\beta)$, and it has the analytical solution

$$(\mathbf{X}^\top \mathbf{X})\hat{\beta} = \mathbf{X}^\top \mathbf{Y}.$$

And if $\mathbf{X}$ has full column rank, then the unique least squares estimator is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

i Calculation

We can proof the above theorem by taking the derivative of $S(\beta)$ with respect to β and setting it equal to zero. This leads to the normal equations, which can be solved for $\hat{\beta}$.

i) Expansion of the criterion

Expanding the quadratic form gives

$$S(\beta) = \mathbf{Y}^\top \mathbf{Y} - 2\beta^\top \mathbf{X}^\top \mathbf{Y} + \beta^\top \mathbf{X}^\top \mathbf{X} \beta.$$

This is a quadratic function of β .

ii) Derivation of the Normal Equations

To minimize $S(\beta)$, differentiate with respect to β :

$$\frac{\partial S(\beta)}{\partial \beta} = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X} \beta.$$

Setting this equal to zero yields the **normal equations**:

$$\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{Y}.$$

These are the equations that define the ordinary least squares estimator.

Full column rank case

If $\mathbf{X}$ has full column rank p , then $\mathbf{X}^\top \mathbf{X}$ is invertible, and the unique least squares estimator is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

This is the ordinary least squares estimator, or OLS estimator.

i Remark – When is this formula valid?

The closed-form expression above requires

$$\text{rank}(\mathbf{X}) = p.$$

This means the columns of $\mathbf{X}$ are linearly independent. If they are not, then $\mathbf{X}^\top \mathbf{X}$ is singular, and special care is needed. We will discuss rank deficiency later.

3 4. Geometric Interpretation

i Recap of Column space of the design matrix

Recall the column space of $\mathbf{X}$:

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\beta : \beta \in \mathbb{R}^p\}.$$

This is the set of all vectors that can be represented by the linear model.

3.1 Least squares as projection

The fitted value vector is

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}.$$

Since $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$, least squares chooses the vector in the column space of $\mathbf{X}$ that is closest to $\mathbf{Y}$ in Euclidean distance.

Thus, $\hat{\mathbf{Y}}$ is the **orthogonal projection** of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$.

The residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X}\hat{\beta}.$$

Geometrically, $\mathbf{e}$ is the part of $\mathbf{Y}$ orthogonal to the model space $\mathcal{C}(\mathbf{X})$.

4 Orthogonality Properties

A central property of least squares is that the residual vector is orthogonal to every column of $\mathbf{X}$.

4.1 5.1 Algebraic proof

Starting from the normal equations,

$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y},$$

we rearrange to obtain

$$\mathbf{X}^\top (\mathbf{Y} - \mathbf{X} \hat{\boldsymbol{\beta}}) = \mathbf{0}.$$

Since $\mathbf{e} = \mathbf{Y} - \mathbf{X} \hat{\boldsymbol{\beta}}$, this becomes

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

Therefore, $\mathbf{e}$ is orthogonal to every column of $\mathbf{X}$.

4.2 5.2 Consequences

This implies:

- the residuals sum to zero if the model includes an intercept;
- fitted values and residuals are orthogonal;
- the least squares fit is a projection onto the model space.

If the first column of $\mathbf{X}$ is $\mathbf{1}$, then

$$\mathbf{1}^\top \mathbf{e} = 0,$$

so

$$\sum_{i=1}^n e_i = 0.$$

5 6. The Hat Matrix

5.1 6.1 Definition

In the full rank case,

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

Define the matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top.$$

Then

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}.$$

The matrix $\mathbf{H}$ is called the **hat matrix** because it puts the “hat” on $\mathbf{Y}$.

5.2 6.2 Properties of the hat matrix

The hat matrix satisfies two important properties:

5.2.1 Symmetry

$$\mathbf{H}^\top = \mathbf{H}.$$

5.2.2 Idempotence

$$\mathbf{H}^2 = \mathbf{H}.$$

A matrix that is both symmetric and idempotent is the matrix of an orthogonal projection.

Thus, $\mathbf{H}$ projects vectors onto $\mathcal{C}(\mathbf{X})$.

5.3 6.3 Residual maker matrix

Define

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

Then the residual vector can be written as

$$\mathbf{e} = \mathbf{MY}.$$

The matrix $\mathbf{M}$ is also symmetric and idempotent, and it projects onto the orthogonal complement of $\mathcal{C}(\mathbf{X})$.

6 7. Sum of Squares Decomposition

Because $\hat{\mathbf{Y}}$ and $\mathbf{e}$ are orthogonal, we have

$$\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$$

with

$$\hat{\mathbf{Y}}^\top \mathbf{e} = 0.$$

Hence,

$$\|\mathbf{Y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\mathbf{e}\|^2.$$

This is a Pythagorean identity.

In regression with an intercept, a more familiar decomposition is

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

Later we will call these:

- total sum of squares (SST),
- regression sum of squares (SSR),
- error sum of squares (SSE).

7 8. Statistical Properties of the OLS Estimator

Assume

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon, \quad \mathbb{E}[\varepsilon] = \mathbf{0}, \quad \text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n.$$

7.1 8.1 Expectation

Using

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y},$$

we get

$$\mathbb{E}[\hat{\beta}] = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbb{E}[\mathbf{Y}] = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \beta = \beta.$$

Thus, OLS is unbiased.

7.2 8.2 Variance

Also,

$$\text{Var}(\hat{\beta}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{Var}(\mathbf{Y}) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}.$$

Since $\text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$, this becomes

$$\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

This formula will be fundamental for confidence intervals and hypothesis tests later.

Consider the simple regression dataset

$$\begin{array}{c|cccc} x_i & 0 & 1 & 2 & 3 \\ \hline y_i & 1 & 3 & 3 & 5 \end{array}$$

Hand calculation

Then

$$\mathbf{Y} = \begin{bmatrix} 1 \\ 3 \\ 3 \\ 5 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}.$$

First compute

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}, \quad \mathbf{X}^\top \mathbf{Y} = \begin{bmatrix} 12 \\ 24 \end{bmatrix}.$$

Hence,

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

Since

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{20} \begin{bmatrix} 14 & -6 \\ -6 & 4 \end{bmatrix},$$

we obtain

$$\hat{\beta} = \frac{1}{20} \begin{bmatrix} 14 & -6 \\ -6 & 4 \end{bmatrix} \begin{bmatrix} 12 \\ 24 \end{bmatrix} = \begin{bmatrix} 1.2 \\ 1.2 \end{bmatrix}.$$

Thus, the fitted line is

$$\hat{Y} = 1.2 + 1.2x.$$

R demonstration

Fit we fit the model


```
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)
df <- data.frame(x = x, y = y)

fit <- lm(y ~ x, data = df)
coef(fit) # check the coefficient
```

```
(Intercept)      x
          1.2      1.2
```

```
X <- model.matrix(fit)
X
```

```
(Intercept) x
1           1 0
2           1 1
3           1 2
4           1 3
attr("assign")
[1] 0 1
```

```
Y <- matrix(y, ncol = 1)
beta_hat <- solve(t(X) %*% X) %*% t(X) %*% Y
beta_hat
```

```
      [,1]
(Intercept) 1.2
x            1.2
```

```
y_hat <- X %*% beta_hat
e <- Y - y_hat

y_hat
```

```
      [,1]
1  1.2
2  2.4
3  3.6
4  4.8
```

```
e
```

```
      [,1]  
1 -0.2  
2  0.6  
3 -0.6  
4  0.2
```

```
H <- X %*% solve(t(X) %*% X) %*% t(X)  
round(H, 4)
```

```
      1  2  3  4  
1  0.7 0.4 0.1 -0.2  
2  0.4 0.3 0.2  0.1  
3  0.1 0.2 0.3  0.4  
4 -0.2 0.1 0.4  0.7
```

```
round(H - t(H), 8)
```

```
      1 2 3 4  
1 0 0 0 0  
2 0 0 0 0  
3 0 0 0 0  
4 0 0 0 0
```

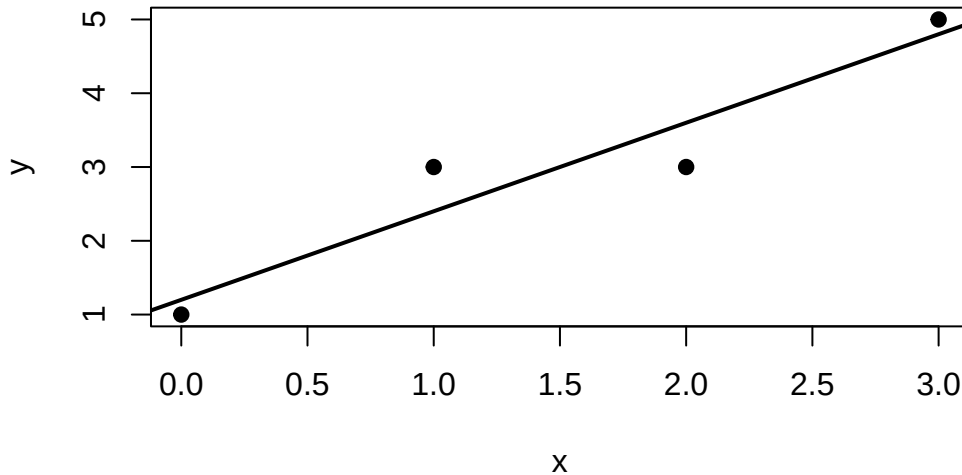
```
round(H %*% H - H, 8)
```

```
      1 2 3 4  
1 0 0 0 0  
2 0 0 0 0  
3 0 0 0 0  
4 0 0 0 0
```

```
round(t(X) %*% e, 8)
```

```
      [,1]  
(Intercept)  0  
x             0
```

```
plot(x, y, pch = 19, xlab = "x", ylab = "y")
abline(fit, lwd = 2)
```



11. Interpretation of the Geometry

The vector $\mathbf{Y}$ lives in $\mathbb{R}^n$. The model space $\mathcal{C}(\mathbf{X})$ is a p -dimensional subspace of $\mathbb{R}^n$ when $\mathbf{X}$ has full rank.

Least squares finds the point in this subspace that is closest to $\mathbf{Y}$.

This is why • fitted values are projections, • residuals are orthogonal to the model space, • sum of squares decompositions are geometric identities.

12. In-Class Discussion Questions

1. Why does minimizing $\|\mathbf{Y} - \mathbf{X}\beta\|^2$ lead to orthogonality?
2. Why is the hat matrix called a projection matrix?
3. Why does the inclusion of an intercept imply that the residuals sum to zero?
4. What fails when $\mathbf{X}$ does not have full column rank?

13. Practice Problems

Conceptual 1. Explain why $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$ is a geometric statement. 2. Give an interpretation of $\mathcal{C}(\mathbf{X})$ in the context of regression. 3. Explain the difference between $\mathbf{H}$ and $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$.

Computational

Let

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 2 \end{bmatrix}, \quad \mathbf{Y} = \begin{bmatrix} 1 & 2 & 2 \end{bmatrix}.$$

1. Compute $\mathbf{X}^\top \mathbf{X}$.
2. Compute $\mathbf{X}^\top \mathbf{Y}$.
3. Find $\hat{\beta}$.
4. Compute $\hat{\mathbf{Y}}$ and $\mathbf{e}$.
5. Verify that $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$.

Proof-based

Show that the hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$$

is symmetric and idempotent.

14. Suggested Homework

Complete the following tasks:

- derive the normal equations from the least squares criterion;
- prove that $\hat{\mathbf{Y}}$ is the projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$;
- prove that $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$;
- verify that $\mathbf{H}$ is symmetric and idempotent;
- fit a simple regression model in R and compute: $\hat{\beta}$, $\hat{\mathbf{Y}}$, $\mathbf{e}$, $\mathbf{H}$.

15. Summary

In this week, we introduced the least squares estimator and showed that it solves the normal equations

$$\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{Y}.$$

When $\mathbf{X}$ has full column rank,

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

Geometrically, least squares is projection onto the column space of $\mathbf{X}$. This leads naturally to the hat matrix, residual orthogonality, and sum of squares decompositions.

Next week, we will study the distribution theory of OLS under the normal error model, including estimation of σ^2 , standard errors, and inference.

Side fact: the Matrix Calculus Facts Used This Week

For a constant vector $\mathbf{a}$, vector β and constant matrix $\mathbf{A}$, we have

i) $\frac{\partial}{\partial \beta}(\mathbf{a}^\top \beta) = \mathbf{a}$, and

ii) if $\mathbf{A}$ is symmetric, $\frac{\partial}{\partial \beta}(\beta^\top \mathbf{A} \beta) = 2\mathbf{A}\beta$.

These identities justify the derivative of the least squares criterion.

References

Montgomery, Douglas C. 2017. *Design and Analysis of Experiments*. John Wiley & Sons.

Montgomery, Douglas C, Elizabeth A Peck, and G Geoffrey Vining. 2021. *Introduction to Linear Regression Analysis*. John Wiley & Sons.

Petersen, Kaare Brandt, and Michael Syskind Pedersen. 2008. *The Matrix Cookbook*. Technical University of Denmark.

Seber, George AF, and Alan J Lee. 2003. *Linear Regression Analysis*. John Wiley & Sons.

Part I

Appendix

Matrix Algebra Review

Matrix dimensions

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times p$, then $\mathbf{AB}$ is defined and is $m \times p$.

Transpose rules

For conformable matrices,

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top.$$

Inverse rule

If $\mathbf{A}$ and $\mathbf{B}$ are invertible, then

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}.$$

Symmetric matrices

A matrix $\mathbf{A}$ is symmetric if

$$\mathbf{A}^\top = \mathbf{A}.$$

Covariance matrices are always symmetric.

A good reference of the linear algebra can be found from Petersen and Pedersen (2008).

- Petersen, Kaare Brandt, and Michael Syskind Pedersen. (2008). *The Matrix Cookbook*. Technical University of Denmark.